

Data Analysis Memo

Hugo Novales

2026-04-18

1. Research Question

What explains variation in majority formation in multiparty legislatures? There is consensus in political science about the role of district magnitude and party fragmentation (Cox 1997; Lijphart 1999; Norris 2004). However, does this rationale extend to majority formation? This question is important because it speaks to the actual performance of legislatures. It is also important because majority formation is how political and policy change occurs in democracies. If majorities are not formed in legislatures, then democracy may stagnate and fail to perform in a manner that is satisfactory for citizens. District magnitude is relatively stable, as electoral rules are a matter of contention. Comparing across cases is difficult because specific countries may have different drivers of majority formation or gridlock.

The causal logic is important. Electoral systems do not in and of themselves produce more parties, but, where multiple parties exist, they may repress the number of parties or fail to do so. District magnitude here is a key variable. Cox (1997) and Sartori before him have found that smaller district magnitudes tend to exert pressure on the number of parties, while larger magnitudes relax this pressure and allow for more proliferation.

The number of parties and their relative sizes (fragmentation, effective number of parties) may be associated to majority formation (Mainwaring 1993; Tsebelis 1995). However, fragmentation does not say anything about whether those parties have substantial reasons to agree on policy. Polarization, therefore, could also influence majority formation. While I'm not interested in the effect of polarization, it is important to consider it in a causal model.

I have chosen to focus my analysis on a single case. The first reason is that this allows me to control for noise associated to national context variation. In other words, there may be a reason why politicians in country X tend to form majorities less frequently than politicians in neighboring country Y. It could be ethnic cleavages, political culture, legislative procedural rules, or other unknown factors. Focusing on a single country allows me to isolate my variables of interest, at least to a larger measure than if I were employing a cross-country analysis.

While we know that restrictive electoral systems -such as those with smaller district magnitude- tend to repress or suppress the number of parties, it is rare that a country reduces its district magnitude. Legislators are typically in charge of electoral reform, and they reasonably understand that reducing district magnitude may act against their probability of reelection (Andrews and Jackman 2005; Negretto and Visconti 2018). Therefore, overwhelmingly, district magnitude tends to increase rather than decrease. This presents a problem for identification, as the effects of electoral system reform may linger into the future. Societal fragmentation or political polarization -or any other idiosyncratic variable that affects majority formation- may increase or decrease organically without changes in the electoral system. A better test for the effects of electoral system change on majority formation is when changes in opposite directions occur close in time. Guatemala is one such case. Since the first democratic election in the current regime in 1986, over the course of ten elections and an equal number of legislatures, district magnitude has changed 5 times, increasing 4 times and decreasing once (Brolo 2012; Brolo 2013). This within-case variation allows me to approximate a causal effect much more than in a country where magnitude has only increased. Furthermore, restricting the analysis to a single case with such a short democratic spell sets a high bar for causality: since the number of observations

is very small, each single observation should be understood to have enough leverage so that changes in the dependent variable attributable to factors other than my explanation should significantly weaken my theory.

2. Causal Model and Hypotheses

My theory is as follows. District magnitude is the single most important variable that influences majority formation. First, it will increase the number of parties -presumably with different views- present in a legislature. Second, in increasing the number of parties and creating a more proportional seat allocation, it will make the legislature a more faithful reflection of polarization in society, making majority formation more difficult the more polarized society is. That is, the effect of district magnitude is sensitive to polarization: when polarization increases, the effect should be stronger. Finally, I expect majority formation to be sensitive to political events. One specific type of political event is elections. In the year following an election, when the legislature is newly integrated, they may be motivated to deliver on campaign promises and party cohesion is expected to be strongest. As the next election year approaches, legislators will be more occupied with campaigning, therefore affecting legislative output. This means that my causal model should control for distance to the prior or next election.

Hypothesis 1

Higher district magnitude is associated with lower frequency of majority formation.

Hypothesis 2

The effect of district magnitude is partially mediated by party fragmentation.

Hypothesis 3

District magnitude interacts with societal polarization to decrease the frequency of majority formation.

Hypothesis 4

Majority formation will be less frequent as the next election approaches.

I have modeled these hypothesis as follows:

Majority formation = district magnitude + party fragmentation + polarization + distance to next election + (polarization * district magnitude)

3. Data

The effect of district magnitude on majority formation would be best tested in a controlled environment by setting up n polities and assigning them different district magnitudes, observing them over a constant period of time and counting the frequency with which voting majorities form in the legislature. This is not possible. Instead, I have chosen a single polity -Guatemala-, which has some variation in district magnitude over time. This is the main independent variable. One way to measure the dependent variable -majority formation- would be to analyze voting records to identify the frequency with which each legislature achieves a majority. I don't have access to voting records, but I was able to scrape the website of the Guatemalan legislature for every piece of legislation approved since 1928. During my period of interest, each piece of legislation is officially coded with a decree number ("número de decreto") which includes a index number ranging from 1 to the number of laws passed in a year, followed by a hyphen and the year in which the legislation was passed. This allows me to count the number of pieces of legislation approved in each year. I am then using legislative output as a proxy for majority formation.

Guatemala experienced regime change several times over the course of the 20th century. However, after the democratic election in 1985, the regime has been reasonably stable, holding legislative elections which are reasonably free and fair every five years until 1995, and every four years thereafter. This allows me to use seat allocation data from each of those elections to build a party fragmentation measure. I have obtained this

data from the V-Party Dataset by the V-Dem project. I have also obtained data on political polarization from the V-Dem database.

Table 1: Districts and District Magnitude in Guatemala, 1985-2025

District	1985	1990	1995	1999	2003	2007	2011	2015	2019	2023
El Progreso	2	2	1	1	1	1	1	1	2	2
Baja Verapaz	2	2	1	1	2	2	2	2	2	2
Zacapa	2	2	1	1	2	2	2	2	2	2
Sacatepequez	2	2	1	2	3	3	3	3	3	3
Santa Rosa	2	3	2	3	3	3	3	3	3	3
Sololá	2	2	1	2	3	3	3	3	3	3
Retalhuleu	2	2	1	2	3	3	3	3	3	3
Izabal	2	3	2	3	3	3	3	3	3	3
Chiquimula	2	3	2	2	3	3	3	3	3	3
Jalapa	2	2	1	2	3	3	3	3	3	3
Totonicapán	2	3	2	3	4	4	4	4	4	4
Petén	2	2	1	2	4	4	4	4	4	4
Jutiapa	3	4	3	3	4	4	4	4	4	4
Chimaltenango	2	3	2	3	5	5	5	5	5	5
Suchitepequez	2	3	2	3	5	5	5	5	5	5
Escuintla	4	5	4	4	6	6	6	6	6	6
Quetzaltenango	5	5	4	6	7	7	7	7	7	7
Quiché	4	5	4	5	8	8	8	8	8	8
San Marcos	6	6	5	8	9	9	9	9	9	9
Alta Verapaz	4	5	4	6	9	9	9	9	9	9
Huehuetenango	5	6	5	7	10	10	10	10	10	10
Central	10	10	9	10	11	11	11	11	11	11
Guatemala	6	7	6	12	19	19	19	19	19	19
National List	25	29	16	22	31	31	31	31	32	32

Table 2: Average District Magnitude by Election in Guatemala, 1985-2025

ELECTION	avg_magnitude
1985	4.166667
1990	4.833333
1995	3.333333
1999	4.708333
2003	6.583333
2007	6.583333
2011	6.583333
2015	6.583333
2019	6.666667
2023	6.666667

Table 3: Legislative Output (n) by Year in Guateamla, 1986-2025

year	n	ELECTION
1986	77	1985
1987	98	1985
1988	79	1985
1989	77	1985
1990	78	1985
1991	80	1990
1992	80	1990
1993	38	1990
1994	69	1990
1995	90	1990
1996	145	1995
1997	135	1995
1998	106	1995
1999	52	1995
2000	98	1999
2001	79	1999
2002	81	1999
2003	53	1999
2004	41	2003
2005	100	2003
2006	43	2003
2007	78	2003
2008	73	2007
2009	40	2007
2010	54	2007
2011	40	2007
2012	36	2011
2013	20	2011
2014	24	2011
2015	17	2011
2016	51	2015
2017	19	2015
2018	25	2015
2019	11	2015
2020	42	2019
2021	20	2019
2022	53	2019
2023	18	2019
2024	35	2023
2025	8	2023

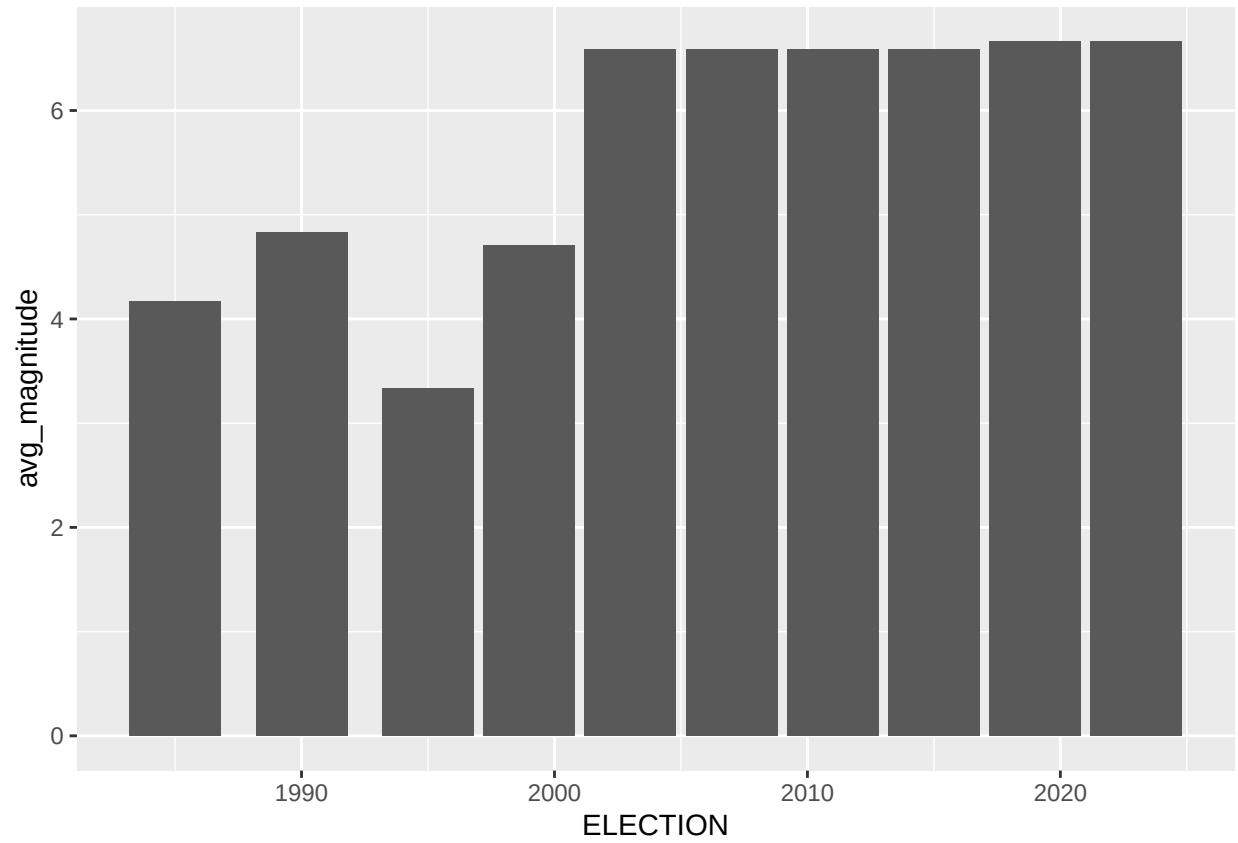


Figure 1: Average District Magnitude by Election

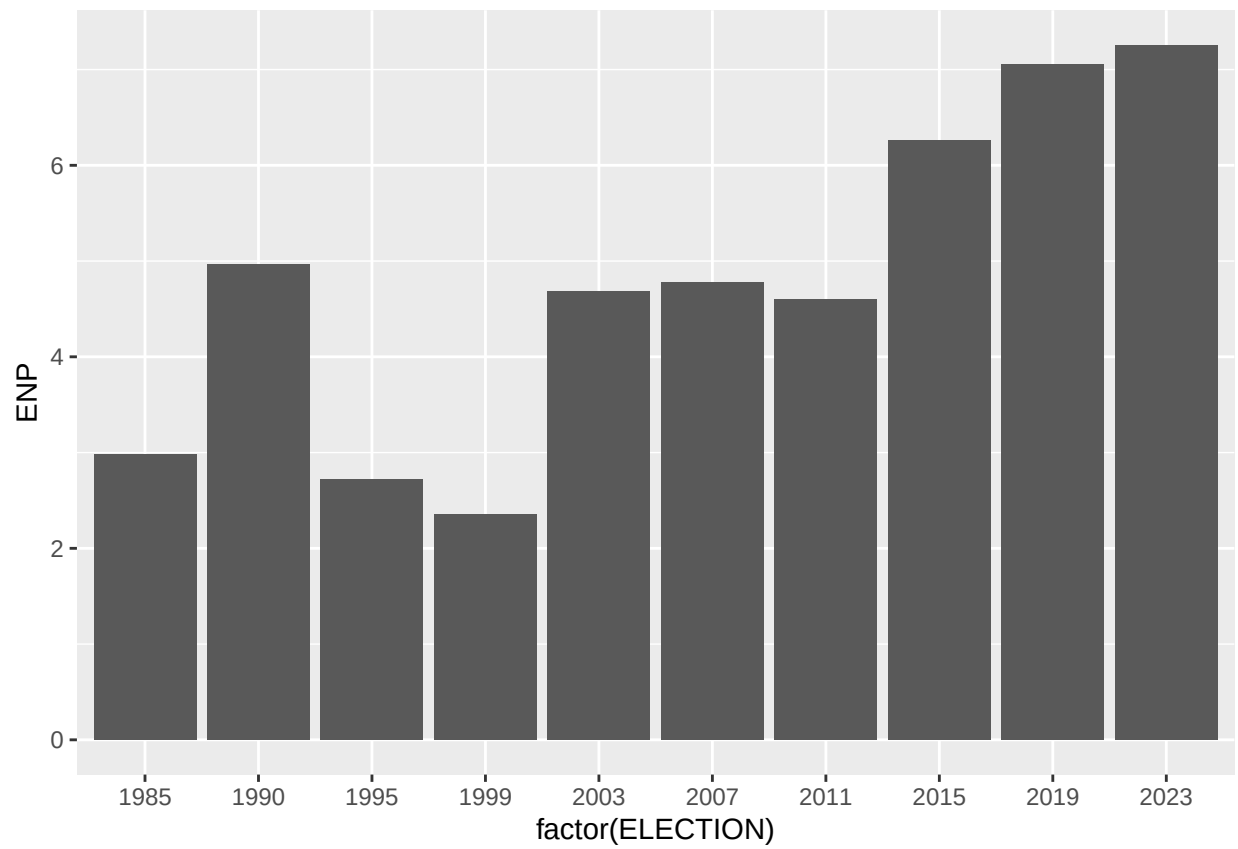


Figure 2: Effective Number of Parties by Election

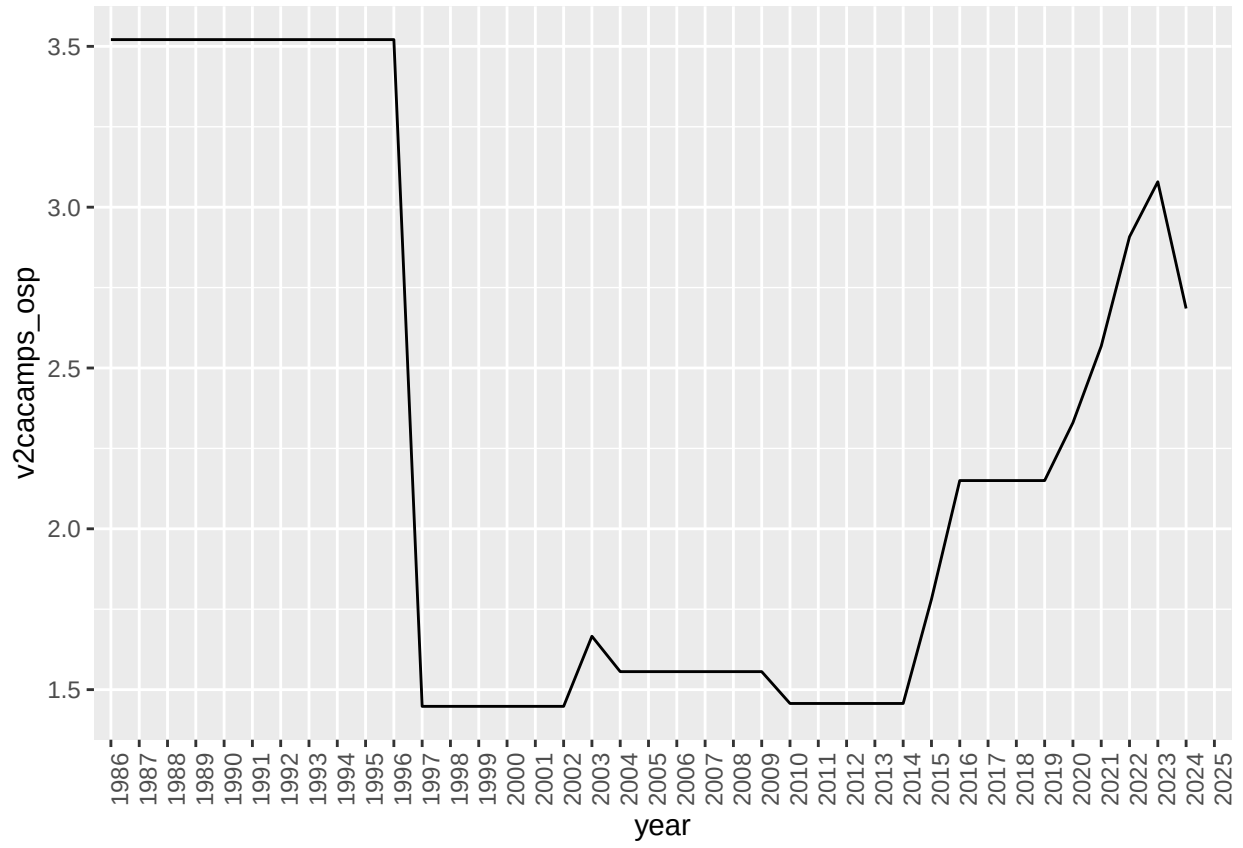


Figure 3: Polarization (v2cacamps_ops) by Year (V-Dem)

```
ggplot(data = decrees_per_magnitude, aes(x = year, y = n)) +  
  geom_col() +  
  facet_wrap(~ELECTION, scales = "free_x")
```

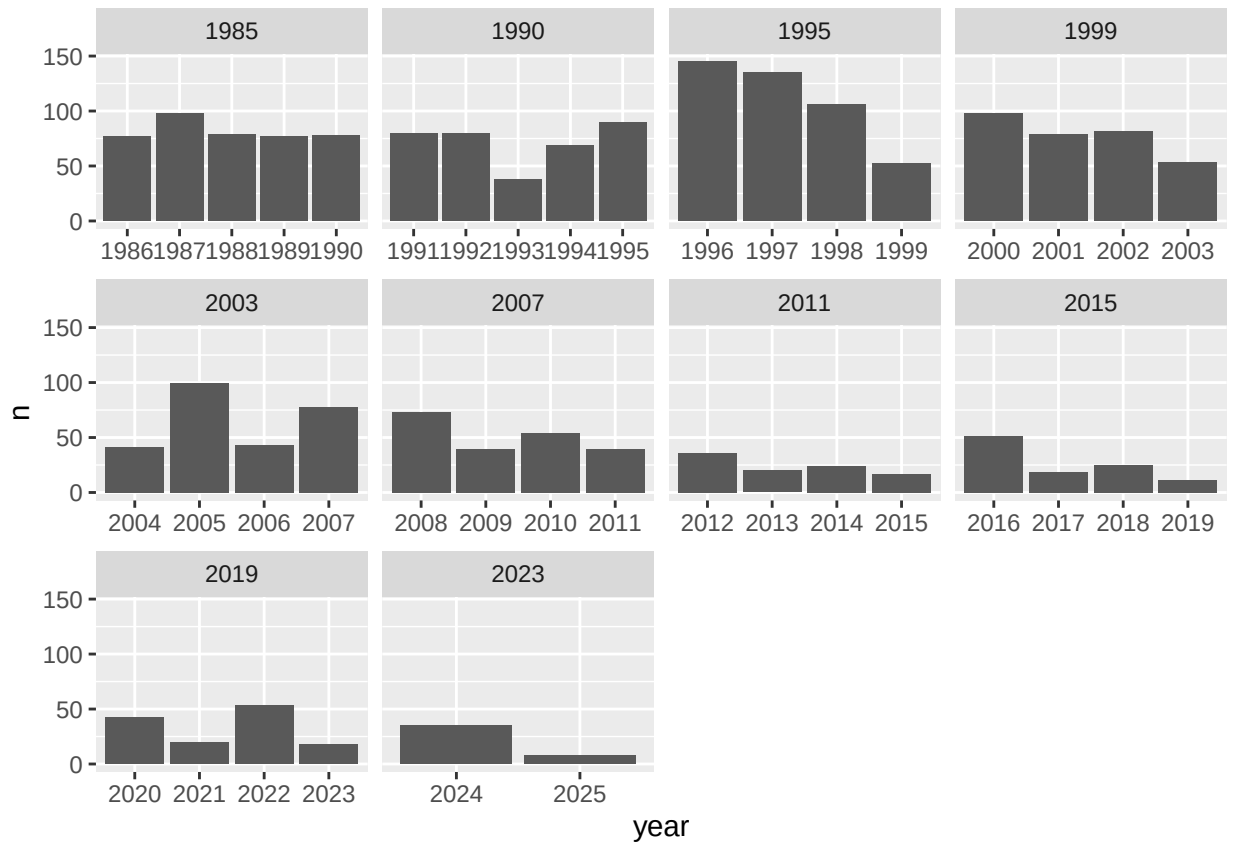


Figure 4: Number of Decrees (n) by Year

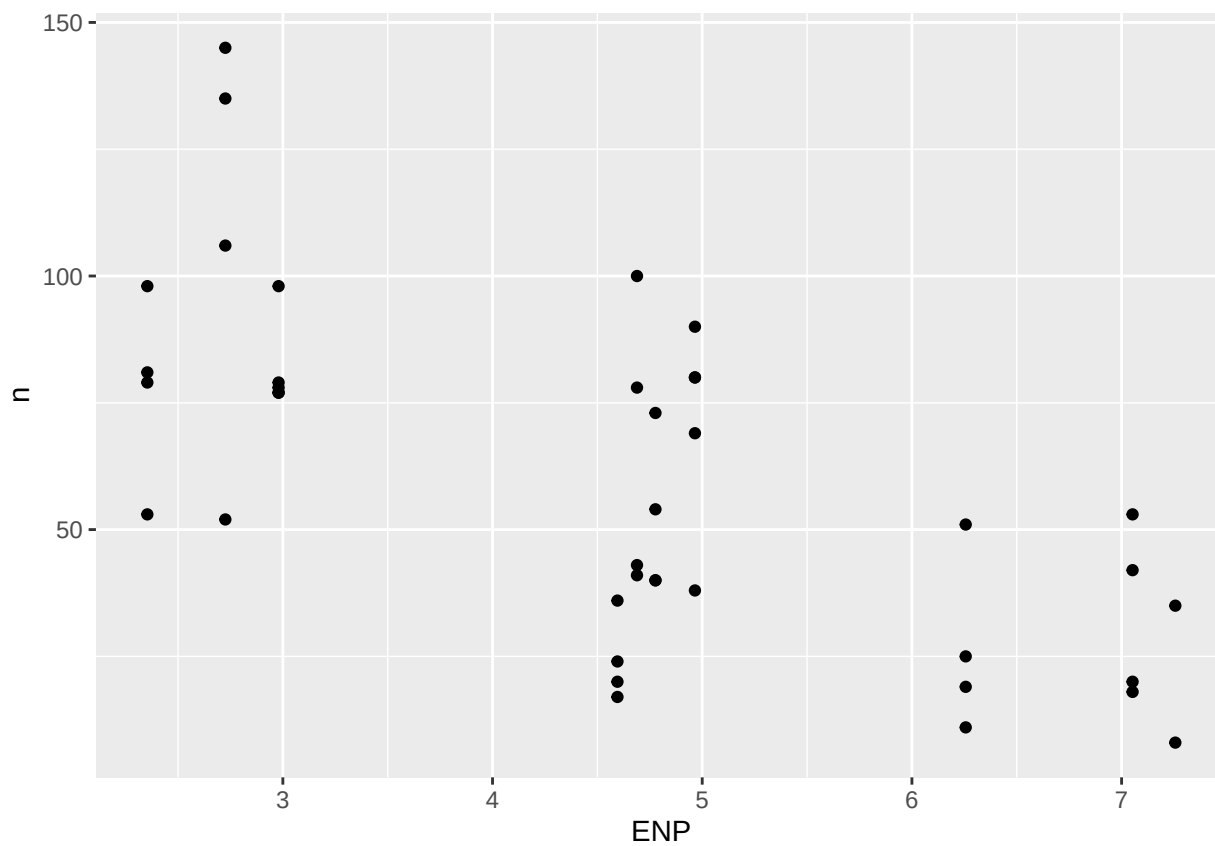


Figure 5: Number of Decrees by Effective Number of Parties

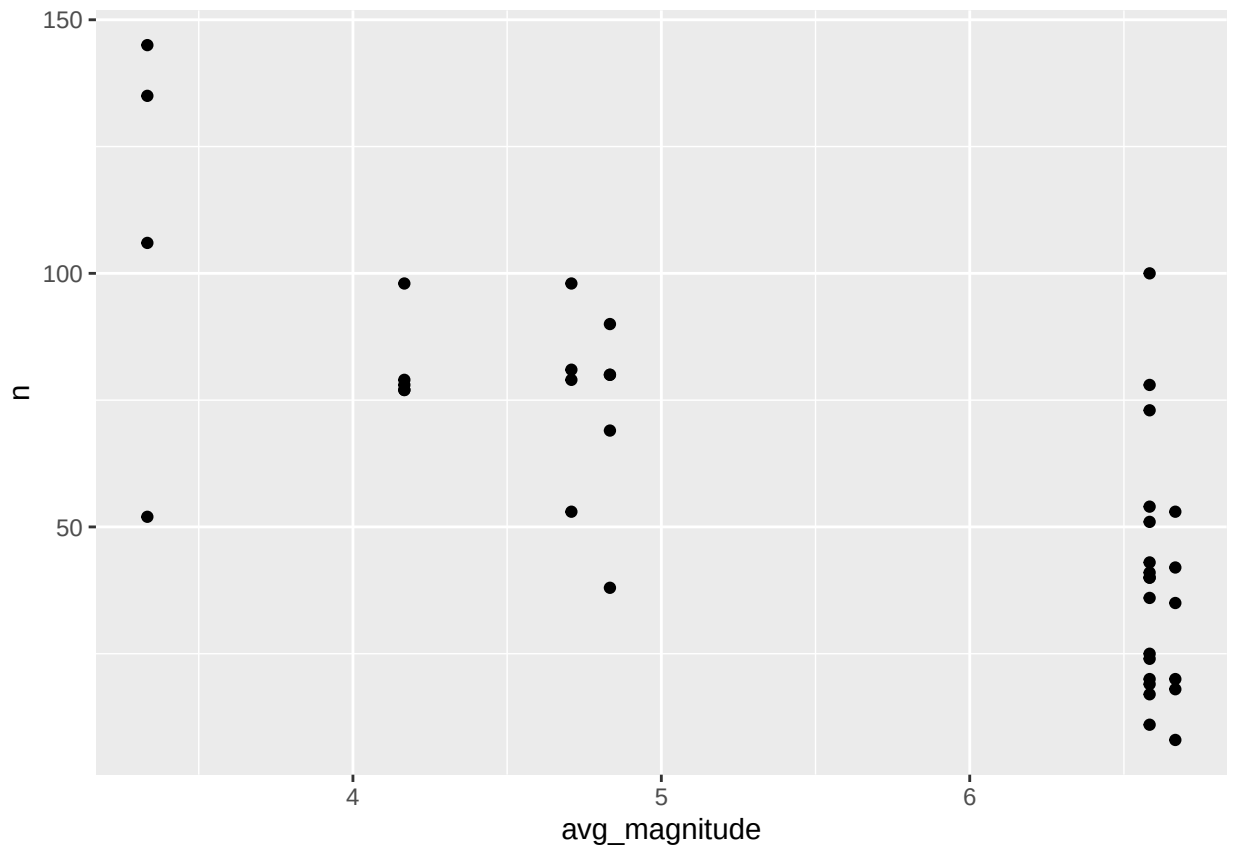


Figure 6: Number of Decrees (n) by Average District Magnitude

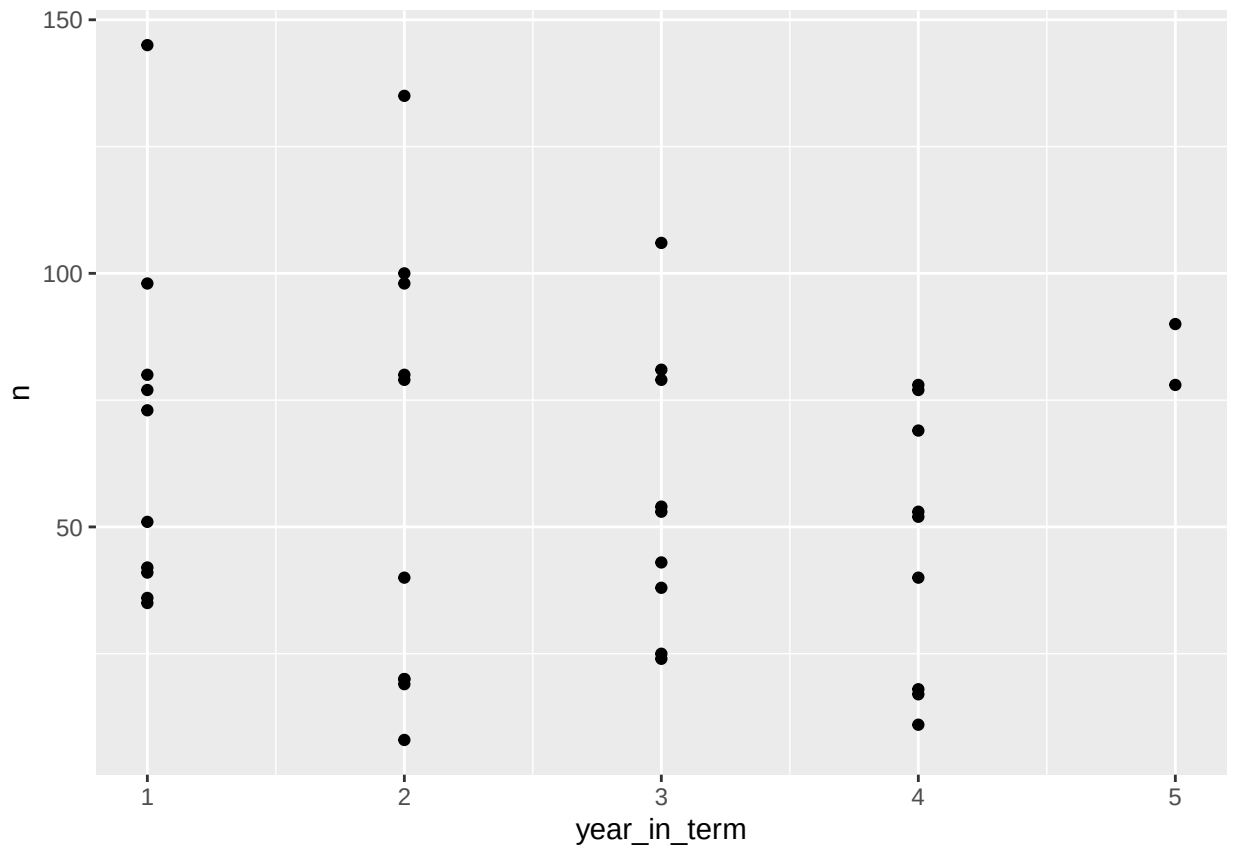


Figure 7: Number of Decrees (n) by Year in Term

Table 4: Results of OLS Regression for Specified Model

	Model 1	Model 2	Model 3	Model 4
(Intercept)	171.191*** (16.798)	190.793*** (18.580)	196.005*** (24.035)	131.173** (47.939)
avg_magnitude	-20.156*** (2.950)	-20.915*** (2.848)	-20.825*** (3.057)	-8.009 (8.780)
year_in_term		-5.973* (2.834)	-6.132* (2.828)	-5.695* (2.787)
v2cacamps_osp			-1.954 (4.271)	27.786 (19.603)
avg_magnitude $\times$ v2cacamps_osp				-6.127 (3.945)
Num.Obs.	40	40	39	39
R2	0.551	0.599	0.601	0.627
R2 Adj.	0.539	0.578	0.567	0.583
AIC	367.2	364.7	356.2	355.5
BIC	372.3	371.5	364.5	365.5
Log.Lik.	-180.618	-178.350	-173.075	-171.739
F	46.679	27.676	17.566	14.309
RMSE	22.12	20.90	20.47	19.78

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

4. Analysis

I ran an OLS regression for my main independent variable -average district magnitude- and my main dependent variable -legislative output- (Model 1), and then I control for year in term and polarization in subsequent models (2 and 3). Based on my model model described above, I included an interaction term where political polarization (v2cacamps_obs) interacts with district magnitude (Model 4). I find that year in term explains a small part of legislative output across all models. In models 1-3, district magnitude is the main driver behind legislative output in the anticipated direction. However, in the last model, the interaction term moves all this explanatory weight to polarization, and significance disappears. Model 2 offers the most parsimonious explanation and is therefore the one I chose to continue working with below. Thus, I find support for Hypothesis 1 and Hypothesis 4, but not for Hypothesis 3.

I then proceed to integrate the Effective Number of Parties (ENP) into the model. As we can see from the results below, the inclusion of ENP significantly reduces the weight of district magnitude. District magnitude is highly correlated with ENP, and we know from conventional literature that district magnitude impacts party fragmentation. Therefore, ENP is assumed here to be a mediating variable and is thus discarded from the model. Thus, I find support for hypothesis 2.

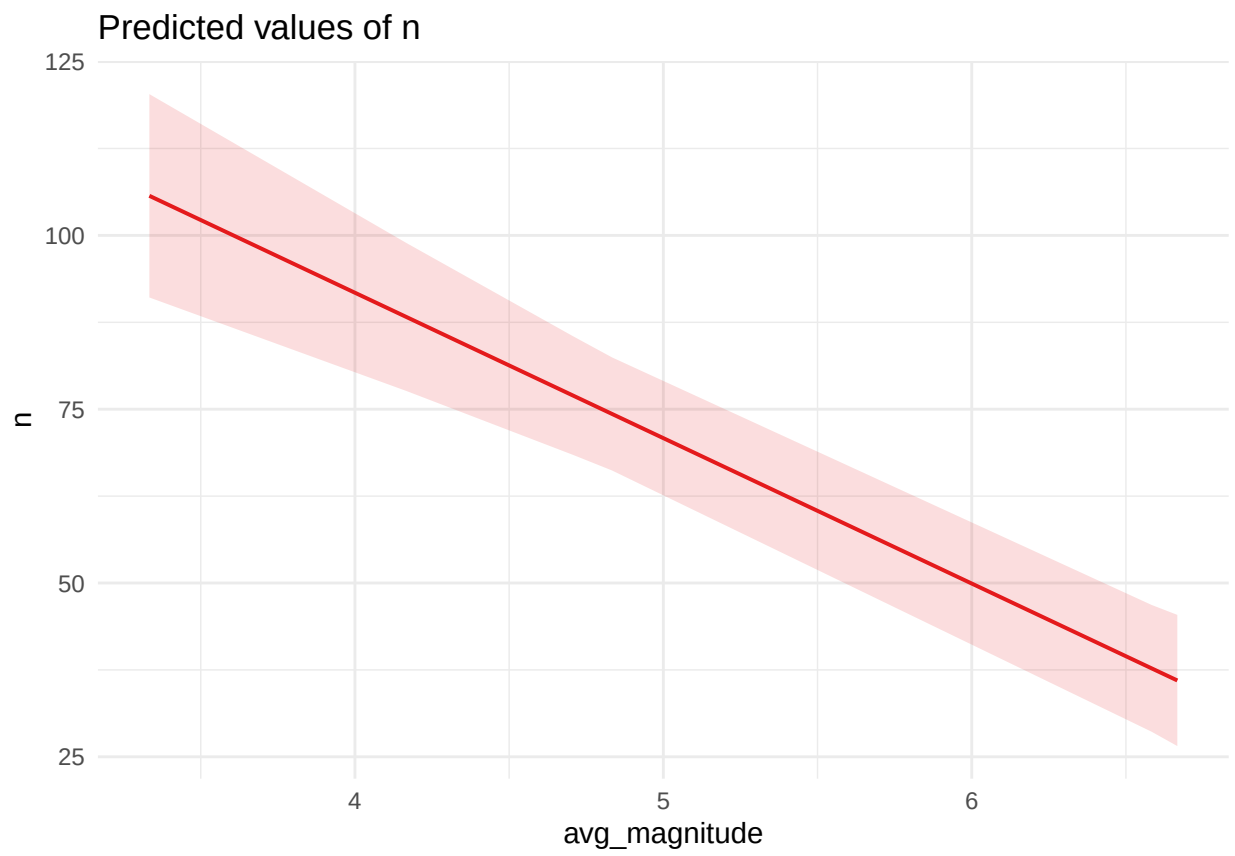


Figure 8: Legislative Output Regressed on Average District Magnitude

Table 5: Results of OLS Regression for Specified Model Controlling for Effective Number of Parties

	Model 1	Model 2	Model 3
(Intercept)	166.571*** (16.944)	186.307*** (18.535)	177.600*** (29.303)
avg_magnitude	-15.235** (4.624)	-15.854*** (4.417)	-14.844* (6.269)
ENP	-4.945 (3.605)	-5.097 (3.438)	-5.154 (4.720)
year_in_term		-6.057* (2.789)	-6.328* (2.826)
v2cacamps_osp			2.037 (5.613)
Num.Obs.	40	40	39
R2	0.573	0.622	0.614
R2 Adj.	0.550	0.591	0.569
AIC	367.3	364.3	356.8
BIC	374.0	372.8	366.8
Log.Lik.	-179.626	-177.165	-172.403
F	24.821	19.780	13.545
RMSE	21.58	20.29	20.12

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

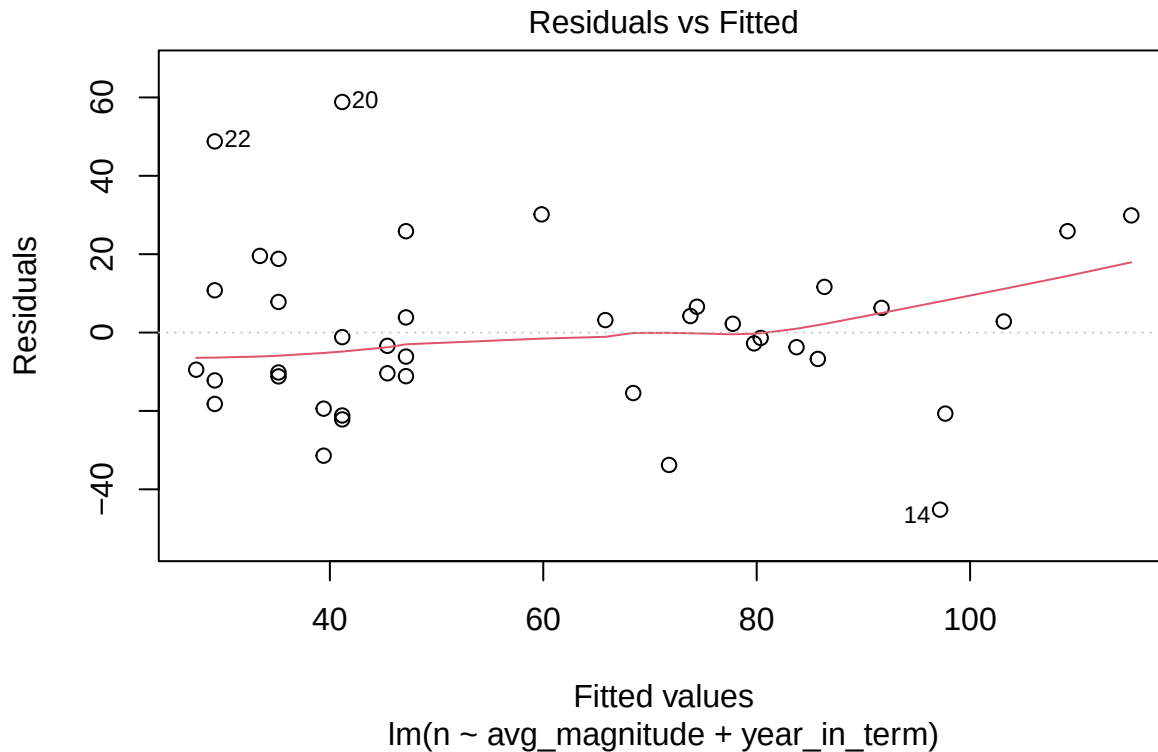
Table 6: Effective Number of Parties Regressed on District Magnitude

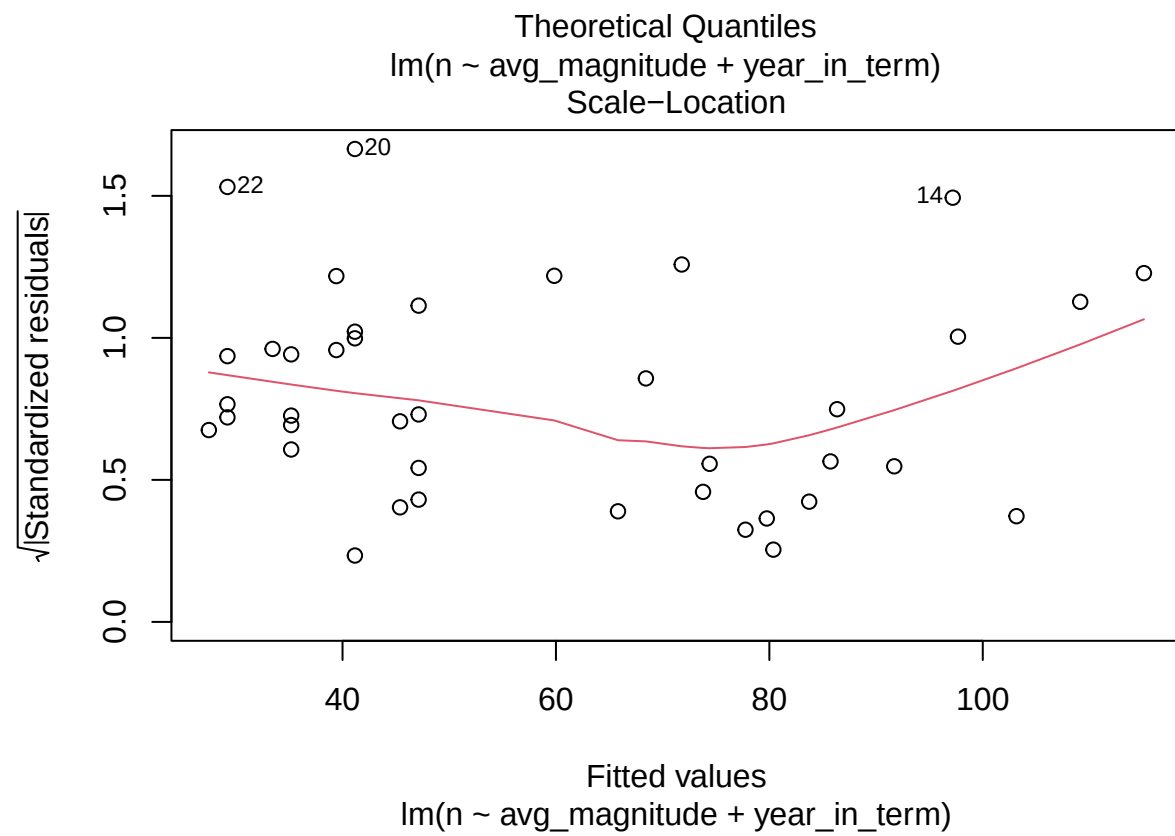
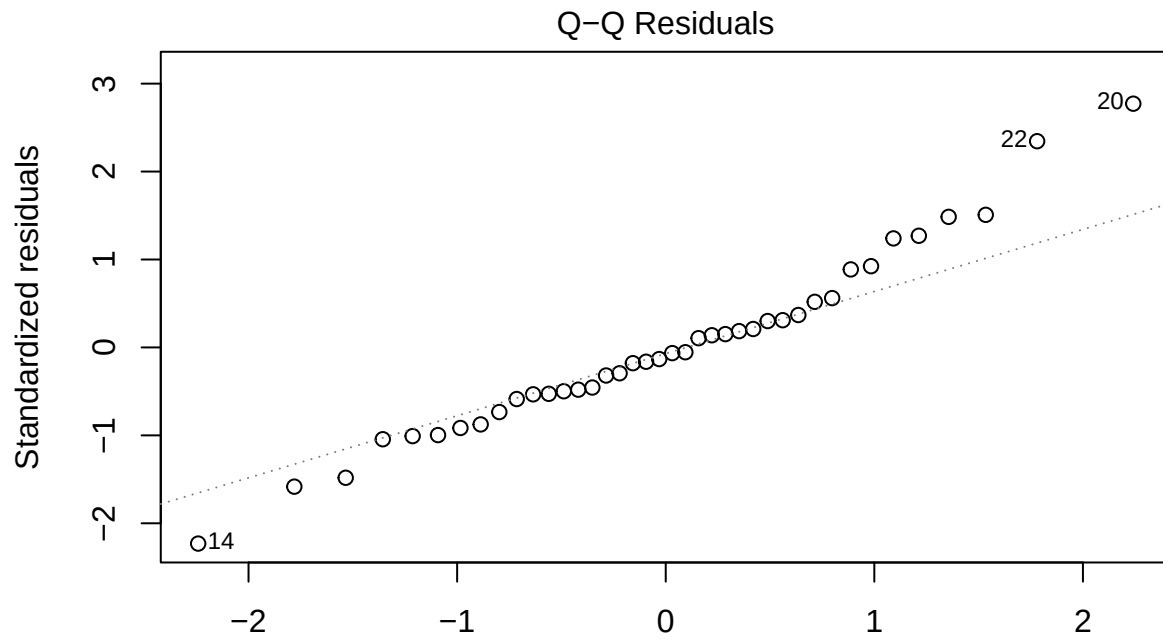
	(1)
(Intercept)	-0.934 (0.747)
avg_magnitude	0.995*** (0.131)
Num.Obs.	40
R2	0.602
R2 Adj.	0.592
AIC	118.2
BIC	123.3
Log.Lik.	-56.110
F	57.504
RMSE	0.98

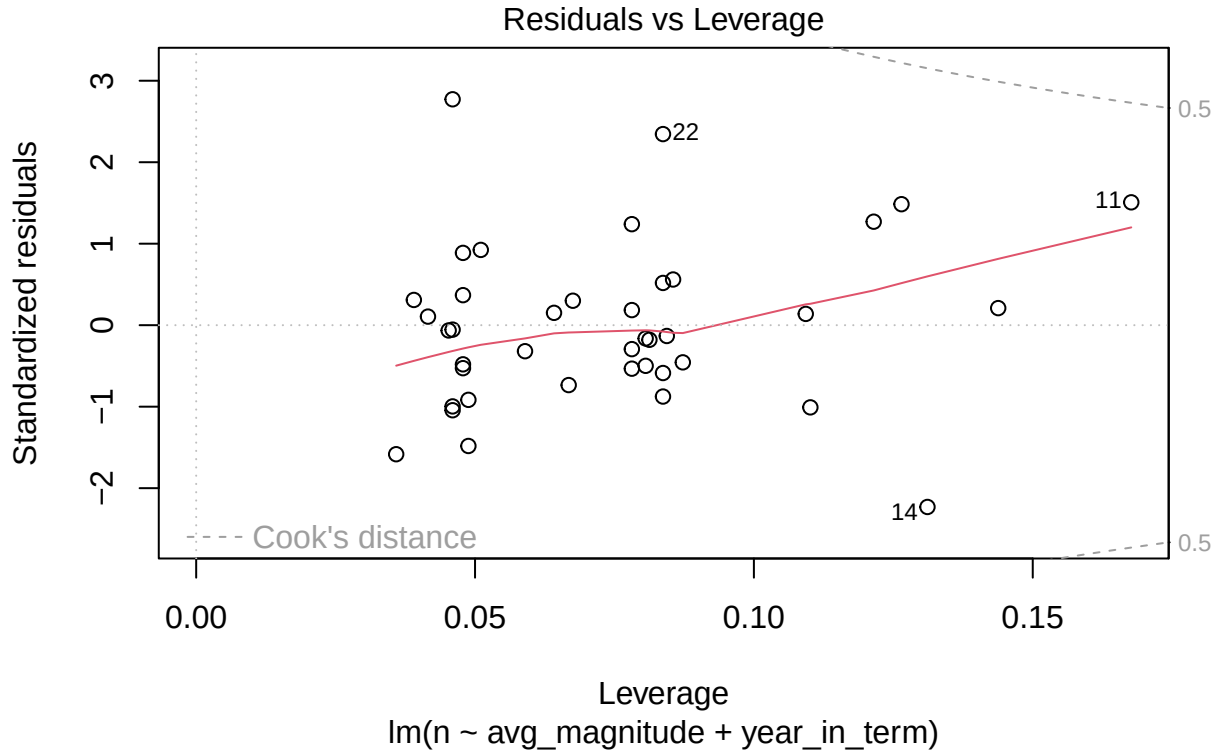
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

5. Assumption Review and Robustness Checks

The following regression diagnostics plots allow me to review the necessary assumptions for my OLS identification strategy. I am using Model 2. The Residuals vs. Fitted plot shows a reasonably stable line and identifies certain outliers (22, 20, 14). The line is slightly curved (U-Shaped) suggesting non-linearity. Similarly, in the Q-Q Residuals plot, while all points lie along the diagonal, the same points as in the Residuals vs. Fitted test lie outside of the diagonal, suggesting that residuals are may not be normally distributed. The Scale-Location plot shows reasonable uniformity in the distribution of all points, suggesting homoscedasticity. Finally, in the Residuals vs. Leverage plot, while the aforementioned outliers have more leverage than the rest of the points, they do not cross into the significance corners.







Since the plots above suggest possible non-linearity and non-normality, I ran statistical tests to check for each of the OLS assumptions. The results are described in Table 7. Overall, Model 2 passes all tests except for the Durbin-Watson test. Observations are not independent, mainly because one important driver of change is the integration of a new legislature with every election. Different legislatures will have different ideological configurations; in some legislatures, the president's party has a majority while in others it does not; also, different issues may come up at different times over 40 years and affect polarization, policy choices and the disposition to form majorities.

Table 8 shows the results for the same models that I specified above, adjusting standard errors to account for clustering around the variable "ELECTION", which defines each legislative session. Coefficients and significance levels remain similar.

Table 7: Assumption Review

Assumption	Test	Statistic	P-Value	Result
Linearity	Rainbow	F = 0.563	0.891	Pass
Independence	Durbin-Watson	DW = 1.526	0.048	Fail
Homoscedasticity	Breusch-Pagan	BP = 0.387	0.824	Pass
Normality	Shapiro-Wilk	W = 0.970	0.367	Pass
Multicollinearity	VIF	avg_magnitude = 1.016, year_in_term = 1.016	—	Pass

```
modelsummary(list ("Model 1" = coeftest1, "Model 2" = coeftest2, "Model 3" = coeftest3, "Model 4" = coeftest4))
```

Table 8: Coefficient Tests for Specified Models on ELECTION Cluster

	Model 1	Model 2	Model 3	Model 4
(Intercept)	171.191*** (13.248)	190.793*** (19.791)	196.005*** (9.617)	131.173** (41.311)
avg_magnitude	-20.156*** (2.920)	-20.915*** (3.082)	-20.825*** (2.305)	-8.009 (9.579)
year_in_term		-5.973+ (3.173)	-6.132+ (3.393)	-5.695+ (2.996)
v2cacamps_osp			-1.954 (3.798)	27.786 (18.158)
avg_magnitude $\times$ v2cacamps_osp				-6.127 (4.113)
Num.Obs.	40	40	39	39
AIC	437.2	430.7	416.2	411.5
BIC	501.4	493.2	474.4	468.0

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

7. Bibliogrpahy

Brolo, Javier. 2012. Resultados electorales de los partidos políticos en Guatemala de 1985 a 2012. Guatemala City: ASIES.

Brolo, Javier. 2013. El tamaño importa: Variaciones por distrito en elecciones legislativas de Guatemala (1984–2011). Guatemala City: ASIES.

Carey, John M., and Simon Hix. 2011. “The Electoral Sweet Spot: Low-Magnitude Proportional Electoral Systems.” *American Journal of Political Science* 55(2): 383–97.

Cox, Gary W. 1997. *Making Votes Count: Strategic Coordination in the World’s Electoral Systems*. Cambridge: Cambridge University Press.

Figueroa, Valentin. 2025. “Does Proportionality Increase Turnout? A Study of Adaptation to Oscillating Electoral Systems.” *Comparative Political Studies OnlineFirst*. <https://doi.org/10.1177/00104140241290209>.

Mainwaring, Scott. 1993. “Presidentialism, Multipartism, and Democracy: The Difficult Combination.” *Comparative Political Studies* 26(2): 198–228.

Negretto, Gabriel L., and Giancarlo Visconti. 2018. “Electoral Reform under Limited Party Competition: The Adoption of Proportional Representation in Latin America.” *Latin American Politics and Society* 60(1): 27–51.

Naslund, Karlerik. 2025. “Regression and Testing Cheat Sheet.” *RPubs*. May 7. https://rpubs.com/knaslundjr/cheat_sheet_2.

Norris, Pippa. 2004. *Electoral Engineering: Voting Rules and Political Behavior*. Cambridge: Cambridge University Press.

Tsebelis, George. 1995. “Decisionmaking in Political Systems: Veto Players in Presidentialism, Parliamentarism, Multicameralism, and Multipartyism.” *British Journal of Political Science* 25(3): 289–325.

Data Sources

Brolo, Javier. 2012. Resultados electorales de los partidos políticos en Guatemala de 1985 a 2012. Guatemala City: ASIES.

Brolo, Javier. 2013. El tamaño importa: Variaciones por distrito en elecciones legislativas de Guatemala (1984–2011). Guatemala City: ASIES.

Congreso de la República de Guatemala. n.d. “Decretos Legislativos.” https://www.congreso.gob.gt/seccion_informacion_legislativa/decretos.

Secretaría General de la Presidencia de la República de Guatemala. 2025. Listado de vetos presidenciales compilado por la Secretaría General, 9 de octubre. Guatemala City: Gobierno de Guatemala. Tribunal Supremo Electoral (TSE). n.d. “Memorias Electorales.” <https://www.tse.org.gt/images/uaip/memorias.electorales.php>.

V-Dem Country-Year+Others Dataset Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Fabio Angiolillo, Michael Bernhard, Agnes Cornell, M. Steven Fish, Linnea Fox, Lisa Gastaldi, Haakon Gjerløw, Adam Glynn, Ana Good God, Allen Hicken, Katrin Kinzelbach, Joshua Krusell, Kyle L. Marquardt, Kelly McMann, Valeriya Mechkova, Juraj Medzihorsky, Anja Neundorf, Pamela Paxton, Daniel Pemstein, Josefine Pernes, Johannes von Römer, Brigitte Seim, Rachel Sigman, Svend-Erik Skaaning, Jeffrey Staton, Aksel Sundström, Marcus Tannenberg, Eitan Tzelgov, Yi-ting Wang, Tore Wig, Steven Wilson and Daniel Ziblatt. 2026. “V-Dem Country-Year Dataset v16” Varieties of Democracy (V-Dem) Project. <https://doi.org/10.23696/vdemds26>.

V-Party Dataset Lindberg, Staffan I., Nils Düpont, Masaaki Higashijima, Yaman Berker Kavasoglu, Kyle L. Marquardt, Michael Bernhard, Holger Döring, Allen Hicken, Melis Laebens, Juraj Medzihorsky, Anja Neundorf, Ora John Reuter, Saskia Ruth–Lovell, Keith R. Weghorst, Nina Wiesehomeier, Joseph Wright, Nazifa Alizada, Paul Bederke, Lisa Gastaldi, Sandra Grahn, Garry Hindle, Nina Ilchenko, Johannes von Römer, Steven Wilson, Daniel Pemstein, Brigitte Seim. 2022. Varieties of Party Identity and Organization (V-Party) Dataset V2. Varieties of Democracy (V-Dem) Project. <https://www.v-dem.net/data/v-party-dataset/>